Research on Multi-View Cattle Facial Recognition Algorithm Based on Feature Matching

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***Abstract*—In recent years, with the advancement of transformation and upgrading of the animal husbandry industry, intelligent and precise breeding technology has gradually become an important direction for the development of modern animal husbandry. Among them, cattle individual identification, as a key link in intelligent farming, is widely used in the fields of health monitoring, reproduction management and behavioural research. In this paper, taking Simmental beef cattle as the research object, we constructed a multi-viewpoint cattle facial dataset and proposed a cattle identification method combining the descriptor-enhanced SuperPoint-Booster feature extraction network with the improved SuperGlue feature matching model. The experimental results show that the model significantly improves the feature point matching accuracy and substantially increases the accuracy of individual cattle recognition under different angles. In the tests on both the small viewpoint variations dataset and the large viewpoint variations dataset, the proposed method outperforms the traditional algorithm combinations in terms of matching accuracy and robustness, with recognition accuracies of 94.7% and 79.5%, respectively. Despite the increase in matching time, the overall performance is significantly improved, demonstrating its vast potential for use in real-world ranch management.**

Keywords—Deep learning, Feature matching, Multi-view Image Processing, cattle identification

I. INTRODUCTION

In recent years, China's cattle farming industry has experienced steady growth, with national policies promoting the transformation and upgrading of the livestock sector. Traditional pasture management methods have become inadequate to meet the demands of modern large-scale farming[1], making precision farming an inevitable trend. In the intelligent management of cattle, individual cattle identification is crucial for tasks such as health monitoring, breeding management, behavioral research[2], and performance tracking[3]. Traditional cattle identification methods are categorized into permanent, semi-permanent, and temporary. Permanent methods, such as "tattoos" and "microchip implants," cause obvious physical harm to the cattle and have poor scalability, making them unsuitable for medium and large-scale farms. Semi-permanent methods like "ear tags" and "identity collars" are widely used but are prone to falling off and damaging the cattle's ears over time. Temporary methods such as RFID and painting suffer from insufficient flexibility.

With the rapid development of deep learning, significant breakthroughs have been made in the field of cattle identification. Chen et al.[4] proposed a deep learning re-identification model called the Global and Partial Network (GPN), which uses three branch modules to extract global and local features from Holstein cow face images as feature representations. An attention mechanism is introduced in the partial branch, and the image is represented as a comprehensive feature vector, which is then used for cow face re-identification, significantly improving recognition accuracy. Meng et al.[5] proposed a cow face recognition method by combining additive margin softmax loss and adversarial reciprocal point learning, which improved recognition accuracy, especially in the case of unknown cow faces, achieving an accuracy rate of 94.46%. Lu et al.[6] successfully identified key regions of cattle in images and effectively extracted features using the LKA module under unsupervised learning conditions. These methods have achieved remarkable results in various cattle identification tasks, not only with high accuracy but also demonstrating good robustness. However, deep learning is costly, requires large amounts of training data, and the frequent turnover and large number of cattle in large farms make model updates expensive, significantly reducing practicality.

Image matching aims to identify and correspond to the same or similar structural content in two or more images. This technology is widely used in visual SLAM, visual tracking, 3D reconstruction, and image retrieval[7]. Image matching technology does not require separate training for newly added individuals, aligning with the concept of intelligent management in large-scale farming, and can be considered a new approach to individual cattle identification. In the field of autonomous driving, image matching technology is widely used for environmental perception, SLAM (Simultaneous Localization and Mapping), and object detection, enhancing real-time scene understanding and target tracking[8]. Aslan et al.[9] used the SURF method to extract features from grayscale images to improve the performance of human activity recognition (HAR), and experiments on the RGB-D dataset showed that this method has a high recognition rate. Weng et al.[10] constructed a new matching recognition model, inputting target cow face data from different angles, and combined the GMS algorithm with the random sample consensus algorithm to achieve accurate individual cattle identification. However, traditional methods such as SIFT[11], SURF[12], and ORB[13] often result in insufficient feature point extraction and low matching accuracy when the cow's face angle changes significantly, making it difficult to handle complex environments." This study focuses on Simmental beef cattle and proposes a multi-angle cattle image matching recognition method, aiming to enrich the theoretical research on cattle face recognition technology. First,

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a descriptor-enhanced SuperPoint-booster[14] self-supervised feature extraction network is used to extract the positions and descriptors of feature points from cow face images. Then, an improved SuperGlue algorithm[15] is used to construct a feature matching model. When performing image matching tasks, this model matches input images from different angles with standard cow front-face images, and finally selects the cattle ID with the most matching points as the final recognition result.

II. Materials and Methods

A. Cattle Facial Dataset

The data used in this study was collected in July 2023 at Shengquan Animal Husbandry Farm in Chifeng City, Inner Mongolia. A total of 3,165 valid videos from 1,055 cattle were captured, with each video lasting between 30 to 50 seconds. After the collection, the raw videos were preprocessed by decomposing them into frames to extract complete individual images. Images with high similarity or poor quality were manually removed. Subsequently, the YOLO object detection algorithm was employed to crop the cattle facial images, resulting in a dataset containing 237,000 facial images from 997 cattle. To validate the performance of the proposed method, facial images from 300 randomly selected cattle were used to construct three datasets based on viewpoint variations: the standard dataset consists of frontal facial images of the cattle, each annotated with a unique cattle ID; the small viewpoint variation dataset includes facial images with viewpoint changes of less than 30°; and the large viewpoint variation dataset contains images with viewpoint changes ranging between 30° and 60°. The structure of the datasets is illustrated in Fig. 1.

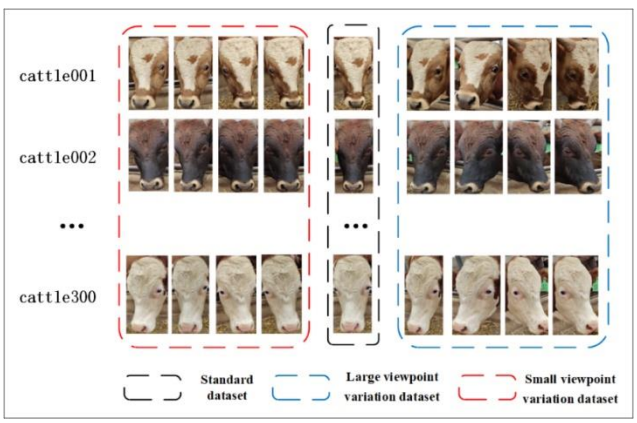

Fig. 1. Dataset structure.

B. Cattle Individual Recognition Model Based on Facial image Feature Matching

1) SuperPoint-booster feature extraction network based on descriptor enhancement.

The design of the SuperPoint feature extraction network draws inspiration from traditional computer vision methods for corner detection and descriptor generation. Its core architecture comprises multiple convolutional layers and pooling layers. By learning to extract features from local receptive fields, SuperPoint effectively identifies salient information in images, such as textures, corner points, and edges, while seamlessly integrating modern deep learning techniques. SuperPoint not only achieves pixel-level precision in extracting feature point locations but also generates corresponding descriptors.

To further enhance the accuracy of feature point descriptors, we introduce a descriptor-enhanced feature-booster network. This network improves the quality of the feature point descriptors, enabling more precise correspondences during the matching process. This is particularly beneficial in scenarios involving low-quality or noisy images, where it effectively reduces matching errors. The integration of the feature-booster network with a dynamic threshold adjustment strategy for non-maximum suppression (NMS) significantly enhances the network's ability to extract and describe feature points. This combination improves feature point matching accuracy under various image conditions and bolsters the network's robustness in handling diverse scenarios.

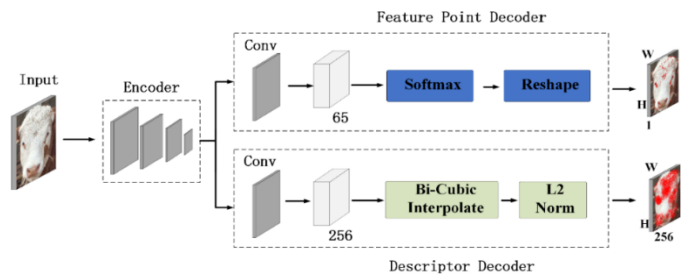

Fig. 2. SuperPoint Feature Point Detection Network Structure Diagram.

Feature-booster is a lightweight network designed to augment the keypoint feature vectors (or descriptors) extracted from an image by an existing keypoint detector, as shown in Fig.3. The network relies only on the feature descriptors and their geometric information (e.g., position, orientation, and scale) as inputs, and outputs new descriptors after enhancement, which can be real-valued vectors or binary vectors. The flow of the method consists of two main steps: self-enhancement and cross-enhancement.

The self-enhancement phase uses an MLP (Multi-Layer Perceptron) to map the original descriptors to the new descriptors. An MLP is first used to approximate the item function, call it MLP_{desc} , and the transformed descriptor d_i^{tr} of the keypoint i is a nonlinear projection of the extracted descriptor d_i :

$$d_i^{tr} < MLP_{ckzc}(d_i) \tag{1}$$

The geometric information is then embedded into the high-dimensional vectors using another MLP to further improve the descriptors. At this point, not only the 2D position of the key point (x_i, y_i) is encoded, but also other information such as the scale S_i , orientation θ_i , and detection score C_i . The high dimensional embedded geometric information is added to the transformed descriptor:

$$d_i^{tr} < d_i^{tr} + MLP_{geo}(p_i) \tag{2}$$

Where $p_i = (x_i, y_i, \theta_i, s_i)$ represents all the available geometric information as described above.

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In the cross-enhancement stage, by inputting N local features in the same image to the Transformer, the enhanced features are output and the attention mechanism in the Transformer is used to aggregate all the local feature information to form a global context. By integrating this global context information, the local feature descriptors can have a larger sensory field and adjust themselves. Feature-booster adopts an Attention-Free Transformer-simple (AFT-simple) instead of the conventional Transformer's Multiple Attention (MHA) operation. AFT rearranges the computation order of Q, K, and V. Attention can be computed by element multiplication instead of matrix multiplication, which makes the time and space complexity linearly related to the context and feature size. The AFT for keypoint B can be expressed as:

$$f_i(X) = \sigma(Q_i) \bigcirc \sum_{j=1}^{N} (soft\max(K) \odot V_j) \tag{3}$$

where $\sigma(\cdot)$ represents the Sigmoid function, Q_i represents row i of Q, and V_j represents row j of V. The Sigmoid function is the same as the Sigmoid function.

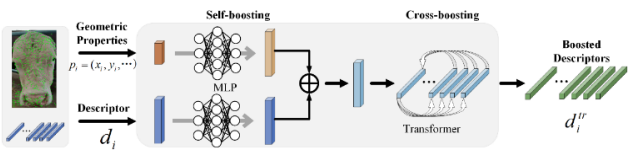

Fig. 3. Feature-booster Network Architecture Diagram

2) Feature Matching Network Based on Improved SuperGlue

The core of SuperGlue's design lies in the integration of the Attentional GNN (Attention Graph Neural Network) and the optimal matching layer. The Attentional GNN enhances the feature representation of key points within a single image using a self-attention mechanism and promotes feature matching between different images through a cross-attention mechanism, enabling the algorithm to learn more advanced and robust features. Meanwhile, the optimal matching layer applies a global optimization strategy to iteratively refine the cost matrix based on the features output by the Attentional GNN, achieving globally optimal matching configurations. However, the self-cross attention mechanism entails high computational complexity, and its cost increases significantly when handling high-resolution images. Additionally, the fixed attention weight update approach can lead to uneven attention distribution in certain scenarios, affecting both matching accuracy and efficiency.

To address these challenges, we propose a convolution-based EfficientAttention module to replace the self-cross attention mechanism in SuperGlue. EfficientAttention incorporates a component named AttnConv, which combines convolutional operations with a context-aware weight generation mechanism. AttnConv significantly reduces computational overhead by aggregating local features through depthwise convolution (DWConv). Moreover, dynamically generated context-aware weights enable the model to adapt flexibly to varying feature distributions across different scenarios, thereby enhancing the ability to capture local details while maintaining computational efficiency.

The enhanced network further optimizes the handling of high-frequency and low-frequency features. High-frequency components are processed using convolution operations to extract local information such as edges and textures, while low-frequency components are preserved through downsampling to maintain global structural information. This approach reduces computational costs while ensuring the integrity of global features. Furthermore, the network incorporates the MemoryEfficientSwish activation function and 1x1 convolutions, which improve feature representation while optimizing computational efficiency. This separation of high- and low-frequency processing strikes a balance between capturing local details and preserving global structures, leading to significant performance improvements in large-scale tasks.

Despite these substantial changes to feature extraction and attention mechanisms, we retain SuperGlue's optimal matching layer to preserve its global matching advantages during the optimization process. Combined with our improved feature extraction and descriptor generation methods, the optimal matching layer continues to leverage its powerful global optimization capabilities to produce accurate keypoint matching results based on the refined features.

III. Experimental Results

Facial images from 300 cattle, along with their corresponding cattle IDs, were saved to construct a standard reference image library. Subsequently, the remaining 2,400 images were used for testing the image matching task. These test images were divided into two datasets: a small viewpoint variation dataset and a large viewpoint variation dataset, each containing 1,200 images. In the small viewpoint variation dataset, the images have viewpoint changes of less than 30°, simulating the capture of cattle faces under relatively stable camera conditions. In contrast, the large viewpoint variation dataset contains images with viewpoint changes ranging between 30° and 60°, simulating scenarios in real pasture environments where unstable camera angles may occur due to changes in camera position or cattle movement. Through this division and setup, the complexity and uncertainty of image acquisition in real pasture environments can be more accurately reflected, thereby better testing the robustness of the image matching algorithm under different viewpoint variations.

C. Comprehensive Evaluation of Algorithm Combination Performance

1) Analysis of the results of the small viewpoint variations

The performance of different algorithm combinations for image matching on the small viewpoint variations dataset is shown in Table 1. The results demonstrate that our proposed model outperforms classical algorithm combinations, achieving significant improvements across multiple key performance metrics. Compared to the original SuperPoint+SuperGlue algorithm, the macro-average precision of our proposed algorithm combination increases from 83.2% to 94.7%. This indicates that our model's filtering algorithm is more effective in eliminating mismatched points in the task of bovine facial image matching, resulting in higher-quality matches.

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Moreover, the macro-average recall improves by 9.4%, reaching 92.1%. This highlights our model's ability to comprehensively identify all possible cattle, thereby reducing the likelihood of omissions. The improvement in the F1-Score further confirms our model's superior ability to maintain a balance between precision and recall, showcasing its robustness in this specific task.

While achieving significant performance improvements, the average matching time of the algorithm combination has not increased substantially, with the average runtime of the original algorithm rising from 0.97 seconds to 1.03 seconds. This indicates a slight increase in the complexity and computational cost of our model.

TABLE I. EXPERIMENTAL RESULTS OF ALGORITHM EVALUATION ON SMALL VIEWPOINT VARIATIONS DATASET

Feature	Matcher	R_{macro}	P_{macro}	$F1_{macro}$	Match time(s)
SIFT	NN	21.6%	27.7%	24.3%	3.31
	NN+ RANSCA	46.9%	48.5%	48,3%	3.93
ORB	NN	46.7%	50.2%	48.5%	**0.70**
	NN+RANSCA	44.0%	41.7%	42.4%	1.58
Dense	LoFTR	86.0%	84.6%	85.1%	1.11
SuperPoint	NN+RANSCA	62.3%	62.3%	62.3%	2.11
	SuperGlue	82.7%	83.2%	81.2%	0.97
	SGMNet	85.4%	86.2%	84.0%	0.89
SuperPoint -booster	NN+RANSCA	65.5%	65.5%	65.5%	1.97
	SGMNet	88.1%	87.3%	86.7%	0.92
	EfficientGlue (Ours)	**92.1%**	**94.7%**	**93.0%**	1.03

The confidence threshold is uniformly set to 0.75 to effectively filter out low-confidence matching points and retain only high-quality matching pairs. By conducting experiments with this strategy, we obtained the feature matching diagram shown in Fig. 4 The figure shows the significant differences in feature matching performance among various algorithms: the matching lines obtained based on SIFT and ORB are sparse and the effect is not ideal; while the feature matching network based on deep learning shows richer and denser matching lines, fully reflecting its advantages in extracting and matching key features of images.

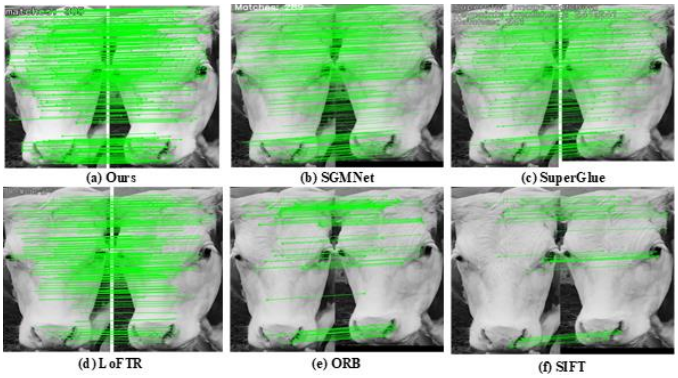

Fig. 4. Matching results of different algorithms

2) Analysis of the results of the large viewpoint variations experiment

As can be seen from the results in the Table 2, the matching performance of various algorithms decreases to different degrees on the large viewpoint variations dataset, and the traditional local feature matching (e.g., SIFT, ORB) falls more significantly. In contrast, deep feature matching methods such as SuperGlue and SGMNet show higher accuracy and F1-score, indicating that deep learning is robust to large viewpoint variations and non-normalised images. It is worth noting that our EfficientGlue significantly outperforms SuperGlue and SGMNet in terms of macro-average accuracy and F1-score, while its average matching time of 1.16s is basically in the same order of magnitude as other deep methods. It can be seen that EfficientGlue can provide better matching accuracy and recall with relatively controllable computational cost in large-view scenarios, achieving a good balance between performance and efficiency.

TABLE II. EXPERIMENTAL RESULTS OF ALGORITHM EVALUATION ON LARGE VIEWPOINT VARIATIONS DATASET

Feature	Matcher	R_{macro}	P_{macro}	$F1_{macro}$	Match time(s)
SIFT	NN	15.2%	16.2%	15.7%	3.21
	NN+ RANSCA	46.6%	43.8%	45.2%	4.05
ORB	NN	25.56%	27.91%	26.68%	**0.71**
	NN+RANSCA	22.82%	25.06%	23.88%	1.64
Dense	LoFTR	74.0%	69.7%	71.1%	1.04
SuperPoint	NN+RANSCA	43.87%	46.67%	45.23%	1.91
	SuperGlue	74.8%	72.8%	72.7%	1.07
	SGMNet	76.0%	75.3%	75.4%	1.03
SuperPoint-booster	NN+RANSCA	45.1%	48.7%	48.6%	2.04
	SGMNet	77.3%	76.1%	76.8%	1.05
	EfficientGlue (Ours)	**81.1%**	**79.5%**	**80.4%**	1.16

Similarly, by setting the confidence threshold uniformly to retain high-quality matching pairs, the matching difficulty has been greatly increased due to the large perspective change. In order to have a clear contrast effect, the threshold is lowered and set to 0.65 to obtain more matching lines, as shown in Fig. 5 Even under low threshold conditions, SIFT and ORB still have only a few pairs of matching points. EifficientGlue shows the most matching pairs, which fully demonstrates that the image matching algorithm based on SuperPoint-booster + EfficientGlue still has excellent processing capabilities even when the matching difficulty increases, and is suitable for individual cattle recognition tasks in various environments.

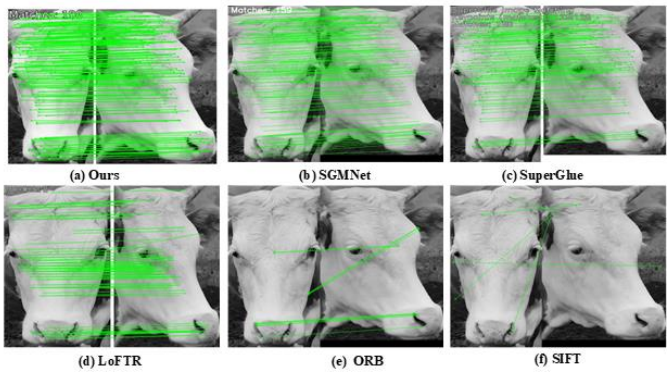

Fig. 5. Matching results of different algorithms

The experimental results mentioned above demonstrate that the proposed algorithm significantly outperforms traditional methods (e.g., SIFT, ORB) as well as existing deep learning approaches (e.g., SuperGlue, SGMNet) across all key

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performance metrics, both on datasets with small and large viewpoint variations.

IV. Conclusion

This study addresses the challenges of individual cattle recognition in large-scale pasture environments, particularly focusing on the difficulties of multi-viewpoint facial image acquisition. We propose a recognition framework that combines a descriptor-enhanced SuperPoint-Booster feature extraction network with an improved SuperGlue feature matching algorithm. This framework was extensively tested on multi-viewpoint cattle facial images collected from real pasture environments. The results demonstrate that, for experimental datasets with small viewpoint variations (<30°) and large viewpoint variations (30°–60°), the proposed method outperforms traditional methods such as SIFT and ORB, as well as feature matching approaches based on SuperGlue and SGMNet, in terms of macro-average precision, macro-average recall, and F1-score. The highest accuracy achieved was 94.7%. Moreover, the method maintains robust recognition performance even under significant viewpoint deviations. Additionally, the feature matching approach eliminates the need for large-scale retraining of deep models when new cattle are added or when there is frequent turnover of cattle, significantly enhancing the practicality and scalability of the system in large-scale pasture environments. Furthermore, the average matching time remains within an acceptable range, laying the foundation for future applications in other livestock species or more complex environments. This work provides important support for the intelligent and information-driven management of modern livestock farming.

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